



**International Collegiate Programming Contest
2026**

Gran Premio de México 2026 - Repechaje

September 12th, 2026

Contest Session

This problem set contains 14 problems; pages are numbered from 1 to 16.

General information

Unless otherwise stated, the following conditions hold for all problems.

Program name

1. Your solution must be called `codename.c`, `codename.cpp`, `codename.java`, `codename.kt`, `codename.py3`, where *codename* is the capital letter which identifies the problem.

Input

1. The input must be read from standard input.
2. The input consists of a single test case, which is described using a number of lines that depends on the problem. No extra data appear in the input.
3. When a line of data contains several values, they are separated by *single* spaces. No other spaces appear in the input. There are no empty lines.
4. The English alphabet is used. There are no letters with tildes, accents, diaereses or other diacritical marks (ñ, Ä, é, Ì, ô, Ü, ç, etcetera).
5. Every line, including the last one, has the usual end-of-line mark.

Output

1. The output must be written to standard output.
2. The result of the test case must appear in the output using a number of lines that depends on the problem. No extra data should appear in the output.
3. When a line of results contains several values, they must be separated by *single* spaces. No other spaces should appear in the output. There should be no empty lines.
4. The English alphabet must be used. There should be no letters with tildes, accents, diaereses or other diacritical marks (ñ, Ä, é, Ì, ô, Ü, ç, etcetera).
5. Every line, including the last one, must have the usual end-of-line mark.
6. To output real numbers, round them to the closest rational with the required number of digits after the decimal point. Test case is such that there are no ties when rounding as specified.

Problem A – Avatar queries

Avatar aang is in trouble. He’s facing a threat bigger than anything he has faced before and he needs the help of former avatars. However talking with his pasts lives, is exhausting so he wants your help to find the correct avatars to contact.

You are given an array $a_1, a_2, \dots, a_n$ where a_i is how helpful the i -th avatar is, and a_n avatar is the latest avatar before Aang. Consider that an avatar could be unhelpful rather than helpful, so he’ll have a negative value.

You’ll have to answer q queries, consisting of a number x indicating, Aang doesn’t want to go further back than the x -th avatar. So starting from avatar $x + 1$, find the maximum possible sum of helpfulness a contiguous subarray of avatars can give. Tradition indicates that Aang must receive help from at least 1 avatar.

Input

The first line contains two integers n and q ($1 \leq n, q \leq 2 \cdot 10^5$).

The second line contains n integers $a_1, a_2, \dots, a_n$ ($-10^9 \leq a_i \leq 10^9$).

Each of the next q lines contains one integer x ($0 \leq x < n$).

Output

For each query, print the maximum help Aang can get, even if it is negative.

Sample input 1	Sample output 1
5 4	6
-2 3 -1 4 -5	6
0	4
1	-5
3	
4	

Notes

For $x = 0$, all avatars are available. For $x = n - 1$, the only available avatar is a_n .

Problem B – Bridges of Koninsberg ii

The kingdom has n cities. Travel is represented by a **directed** cost matrix M : if $M[i][j] = -1$, travelling directly from city i to city j is impossible; otherwise $M[i][j]$ is its non-negative toll. The matrix need not be symmetric. Also, $M[i][i] = 0$.

Then m meteor updates occur. An update gives two cells (i, j) and (x, y) and an increment k . It adds k to every existing off-diagonal entry in the axis-aligned rectangle whose row range is $[\min(i, x), \max(i, x)]$ and column range is $[\min(j, y), \max(j, y)]$. Entries equal to -1 remain -1 , and diagonal entries remain 0.

After all updates, answer q minimum-cost directed-path queries. For each pair (a, b) , output the minimum total toll from a to b , or -1 if no directed path exists.

Input

The first line contains n ($1 \leq n \leq 750$). Each of the next n lines contains n integers $M[i][j]$ ($-1 \leq M[i][j] \leq 10^6$). The diagonal entries are 0; -1 denotes no directed edge.

The next line contains m ($0 \leq m \leq 10^5$). Each of the next m lines contains i, j, x, y, k ($1 \leq i, j, x, y \leq n, 0 \leq k \leq 10^9$), describing an update as above.

The next line contains q ($1 \leq q \leq 10^5$). Each of the next q lines contains a, b ($1 \leq a, b \leq n$).

Output

Print q lines. The t -th line must contain the minimum cost of a directed path for the t -th query after all updates, or -1 if no such path exists.

Sample input 1	Sample output 1
3	5
0 2 -1	4
2 0 4	9
-1 1 0	
1	
1 1 2 2 3	
3	
1 2	
2 3	
1 3	

Notes

Updates affect matrix entries, not already-computed shortest paths. A missing edge (-1) is never created by an update. A query from a city to itself has answer 0. All final answers fit in a signed 64-bit integer.

Problem C – Costly Roads

There are n vertices connected by $n - 1$ roads. Ignoring direction, the roads form a tree.

For every road, traveling in its two directions may have different costs. An input line $u\ v\ a\ b$ means that traveling from u to v costs a , while traveling from v to u costs b .

For every possible starting vertex r , find the sum of the costs of the unique directed trips from r to all vertices. The trip from r to itself has cost zero.

Input

The first line contains an integer n ($1 \leq n \leq 2 \cdot 10^5$).

Each of the next $n - 1$ lines contains four integers u, v, a , and b ($1 \leq u, v \leq n, u \neq v, 1 \leq a, b \leq 10^6$). The undirected edges (u, v) form a tree. The costs of directions $u \rightarrow v$ and $v \rightarrow u$ are a and b , respectively.

Output

Print n integers. The r -th integer must be the sum of travel costs from vertex r to every vertex.

Sample input 1 1	Sample output 1 0
Sample input 2 3 1 2 4 7 2 3 2 5	Sample output 2 10 9 17
Sample input 3 4 2 1 3 8 2 3 4 6 4 2 5 2	Sample output 3 30 9 23 22

Notes

All required sums fit in signed 64-bit integers.

Problem D – Disco Elysium

Revachol, Jamrock, Precinct 41. The body still hangs behind the Whirling-in-Rags, and the RCM demands paperwork.

Your evidence ledger has N entries, each stamped with a non-negative case number. LOGIC points out the obvious: the Coalition’s clerks do not read your report — they only measure its **Completeness Index**, defined as the smallest non-negative integer that does **not** appear among the case numbers in the ledger. The higher the index, the more thorough you appear.

INLAND EMPIRE whispers that ink is negotiable. You may **overwrite** the case number of at most K entries, replacing each with any non-negative integer you like. You cannot add entries and you cannot tear any out — Kim would notice.

Determine the maximum Completeness Index your ledger can attain.

Input

The first line contains two integers N and K ($1 \leq N \leq 2 \cdot 10^5, 0 \leq K \leq N$) — the number of entries in the ledger and the number of entries you may overwrite.

The second line contains N integers $a_1, a_2, \dots, a_N$ ($0 \leq a_i \leq 10^9$), where a_i is the case number stamped on the i -th entry.

Output

Print a single integer: the maximum Completeness Index achievable by overwriting at most K entries.

Sample input 1 6 2 0 2 3 7 1 9	Sample output 1 6
Sample input 2 4 1 5 5 5 5	Sample output 2 1

Notes

The Completeness Index of a collection of non-negative integers is the smallest non-negative integer absent from it. For example, the index of $\{0, 1, 3\}$ is 2, the index of $\{1, 2\}$ is 0, and the index of $\{\}$ is 0.

Problem E – Expected Snaps

Thanos has an array whose length is a power of two. Before he sees it, every element is chosen independently and uniformly from the integers $1, 2, \dots, K$.

An array is sorted if it is nondecreasing. While the current array is not sorted, Thanos performs a snap:

- with probability $1/2$, he keeps the left half and discards the right half;
- with probability $1/2$, he keeps the right half and discards the left half.

All snap choices are independent of each other and of the initial array. After each snap, he checks the remaining array again. An array of length one is always sorted.

The initial length is $N = 2^n$. Find the expected number of snaps, where the expectation is taken over both the random initial array and all snap choices.

Input

The only line contains two integers n and K – the exponent of the initial length and the number of possible values. The value of n is between 0 and 200000, inclusive, and K is between 1 and 200000, inclusive.

It is guaranteed that nK is at most 20000000.

Output

Let the expected number of snaps be the rational number p/q , where p and q are coprime. Print the residue modulo 998244353 of the product of p and q^{-1} , where q^{-1} is the multiplicative inverse of q modulo 998244353. It is guaranteed that this inverse exists.

Sample input 1 1 2	Sample output 1 748683265
Sample input 2 0 1	Sample output 2 0
Sample input 3 4 5	Sample output 3 280149939

Notes

For $n = 1$ and $K = 2$, the only unsorted initial array is $[2, 1]$. It requires one snap, so the expectation is $1/4$.

Problem F – Following the Mentor

The training committee keeps its n contestants in a single mentoring tree. Contestant 1 is the head coach, and every other contestant i is mentored by exactly one contestant p_i . Contestant i has solved a_i problems so far.

The **team** of a contestant u is u together with everybody whose chain of mentors passes through u : the contestants mentored by u , the ones mentored by those, and so on. Every contestant belongs to their own team.

The season is long and the committee keeps rearranging the tree. There are q events, each of one of four kinds.

- 1 $u\ v$: contestant u is reassigned to be mentored by contestant v , and the whole team of u moves along. If v belongs to the team of u , the reassignment would put a contestant above their own mentor, so the committee **rejects the request and nothing changes**; the same happens when u and v are the same contestant.
- 2 $u\ x$: every member of the team of u solves x additional problems.
- 3 u : report how many problems the team of u has solved in total.
- 4 u : the **mentoring distance** of a member w of the team of u is the number of mentoring links that lead from w up to u . Report the sum of the mentoring distances of every member of the team of u .

In the last two events u counts as a member of its own team, contributing a_u problems and a mentoring distance of 0.

Input

The first line contains two integers n and q ($1 \leq n \leq 10^5$, $1 \leq q \leq 10^5$) — the number of contestants and the number of events.

The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^6$) — the problems each contestant has already solved.

The third line contains $n - 1$ integers $p_2, p_3, \dots, p_n$ ($1 \leq p_i \leq n$) — the mentor of each contestant other than the head coach. The mentoring links form a rooted tree: following mentors from any contestant reaches contestant 1. A mentor may have a larger index than the contestant they mentor. This line is empty when $n = 1$.

Each of the next q lines describes one event and starts with its kind.

- 1 $u\ v$ ($1 \leq u, v \leq n$) — reassign contestant u to mentor v .
- 2 $u\ x$ ($1 \leq u \leq n$, $1 \leq x \leq 10^6$) — everyone in the team of u solves x more problems.
- 3 u ($1 \leq u \leq n$) — total problems solved by the team of u .
- 4 u ($1 \leq u \leq n$) — total mentoring distance inside the team of u .

Output

For every event of kind 3 or 4, print one line with the requested value, in the same order as the events appear in the input.

The answers can exceed 2^{31} . They are compared as a sequence of integers.

Sample input 1 7 6 10 20 30 40 50 60 70 1 1 1 2 2 4 3 2 4 1 2 2 5 3 2 1 2 4 4 1	Sample output 1 130 9 145 12
Sample input 2 5 6 1 2 3 4 5 1 1 2 2 1 2 5 4 1 2 2 10 3 1 1 2 3 4 1	Sample output 2 6 45 9
Sample input 3 3 4 5 5 5 1 2 4 1 1 3 1 4 1 3 3	Sample output 3 3 2 5

Problem G – Grand Rainbow Railway

There are n cities and m proposed undirected railway tracks. The cities are numbered from 1 to n , and the tracks are numbered from 1 to m in input order. Each track has one of $n - 1$ permit colors, numbered from 1 to $n - 1$.

Choose exactly $n - 1$ distinct tracks such that they form a spanning tree and every permit color is used exactly once. A spanning tree is a connected, acyclic graph containing all n cities.

Input

The first line contains two integers n and m ($2 \leq n \leq 120$, $n - 1 \leq m \leq 2500$).

Each of the next m lines contains three integers u , v , and c ($1 \leq u, v \leq n$, $u \neq v$, $1 \leq c < n$), describing an undirected track between cities u and v with permit color c .

Parallel tracks are allowed, even with the same endpoints, provided that no two input lines describe the same unordered pair of endpoints with the same color. Every color from 1 to $n - 1$ occurs in at least one track.

Output

If no valid selection exists, print -1 .

Otherwise, print $n - 1$ distinct integers: the indices of tracks in any valid selection, in any order. Any valid answer is accepted.

Sample input 1	Sample output 1
4 5 1 2 1 2 3 2 3 4 3 1 3 1 2 4 2	1 2 3

Notes

The selected tracks must satisfy both requirements simultaneously: they must connect all cities without a cycle, and their colors must be pairwise distinct. Because there are exactly $n - 1$ selected tracks and exactly $n - 1$ colors, pairwise distinct colors are equivalent to using every color exactly once.

Problem H – Hlaalu’s Ebony

Word has reached House Hlaalu of a merchant blessed with a satchel of **Almsivi Intervention** scrolls — the Tribunal’s gift that lets one step instantly to the nearest temple anywhere on Vvardenfell. No longer bound to the silt strider’s fixed path, the merchant may appear at **any** settlement, in **any** order, to trade raw ebony.

At each of the N settlements, the price of raw ebony is known. The merchant will buy one unit at a settlement and sell it at a (possibly identical) settlement. Since travel order is unrestricted, the maximum profit is the largest selling price minus the smallest buying price. If all prices are equal, the answer is 0.

Given the prices at all N settlements, determine the maximum profit the merchant can achieve.

Input

The first line contains a single integer N ($2 \leq N \leq 2 \cdot 10^5$), the number of settlements on Vvardenfell where raw ebony is traded.

The second line contains N integers $p_1, p_2, \dots, p_N$ ($1 \leq p_i \leq 10^9$), where p_i is the price of one unit of raw ebony at the i -th settlement.

Output

Print a single integer: the maximum profit the merchant can achieve.

Sample input 1 5 7 1 5 3 6	Sample output 1 6
Sample input 2 3 10 10 10	Sample output 2 0

Problem I – Inner Product II

You are given a tree with n vertices. Vertex i has a positive integer weight a_i .
You want to choose positive integers $b_1, b_2, \dots, b_n$.
Each edge of the tree contains one of the characters $<$, $=$, or $>$. If an edge is written as $u \ v \ c$, then:

- if c is equal to $<$, you must have $b_u < b_v$;
- if c is equal to $=$, you must have $b_u = b_v$;
- if c is equal to $>$, you must have $b_u > b_v$.

Among all valid assignments, minimize

$$a_1b_1 + a_2b_2 + \cdots + a_nb_n.$$

For the given constraints, it can be proved that the optimal assignment is unique.

Input

The first line contains one integer n ($2 \leq n \leq 200'000$).
The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 1'000'000$).
Each of the next $n - 1$ lines contains two integers u_i, v_i and one character c_i ($1 \leq u_i, v_i \leq n$, $u_i \neq v_i$, c_i is one of $<$, $=$, $>$), meaning that the tree contains an edge between u_i and v_i , and the required relation is $b_{u_i} \ c_i \ b_{v_i}$.

Output

Print the minimum possible value of

$$a_1b_1 + a_2b_2 + \cdots + a_nb_n$$

in the first line.
In the second line, print the unique optimal assignment $b_1, b_2, \dots, b_n$.
It can be proved that, for the given constraints, the minimum value always fits in a signed 64-bit integer.

Sample input 1 6 5 1 4 3 2 6 1 2 < 2 3 > 2 4 < 4 5 = 4 6 >	Sample output 1 32 1 2 1 3 3 1
Sample input 2 7 9 2 8 1 7 3 6 1 2 = 2 3 < 3 4 > 2 5 = 5 6 < 5 7 >	Sample output 2 76 2 2 3 1 2 3 1

Notes

The first sample requires:

- $b_1 < b_2$,
- $b_2 > b_3$,
- $b_2 < b_4$,
- $b_4 = b_5$,
- $b_4 > b_6$.

One optimal assignment is 1 2 1 3 3 1, and its cost is

$$5 \cdot 1 + 1 \cdot 2 + 4 \cdot 1 + 3 \cdot 3 + 2 \cdot 3 + 6 \cdot 1 = 32.$$

Problem J – Jalisco's Hydration Stations

Etienne and Christian are organizing a Sunday event along Avenida Vallarta, modeled as a straight line. Participant i waits at coordinate x_i .

They must place exactly k hydration stations. Each station may be placed at any real coordinate, and multiple stations may be placed at the same coordinate. After the stations are placed, each participant walks to a station minimizing the absolute distance from their position. If several stations are equally near, choosing any of them gives the same walking distance.

Find the minimum possible sum of the walking distances of all participants.

Input

The first line contains two integers n and k ($1 \leq k \leq n \leq 100000$, $k \leq 25$).

The second line contains n integers $x_1, x_2, \dots, x_n$ ($0 \leq x_1 \leq x_2 \leq \dots \leq x_n \leq 10^9$). Equal coordinates are allowed.

Output

Print one integer: the minimum possible total walking distance.

Although station coordinates may be real numbers, an optimal solution can place every station at a participant coordinate, so the answer is an integer. The answer fits in a signed 64-bit integer but may exceed the 32-bit range.

Sample input 1	Sample output 1
6 3 0 2 3 10 11 20	4

Notes

In the sample, stations can be placed at coordinates 2, 10, and 20. The walking distances are 2, 0, 1, 0, 1, 0, for a total of 4.

Requiring exactly k stations never increases the optimum: any redundant station may be placed at the same coordinate as another station.

Problem K – Kaleidoscopic Talavera

An artisan paints a mural using n axis-aligned rectangular stencils. Rectangle i is the half-open set $[x_{1,i}, x_{2,i}) \times [y_{1,i}, y_{2,i})$. Rectangles may overlap, share boundary segments, or coincide.

A point is covered exactly once if it belongs to exactly one rectangle. The union consists of all points covered by at least one rectangle. Its perimeter is the total length of the boundary separating covered points from uncovered points; shared boundaries inside the union do not contribute.

Find (1) the area covered exactly once and (2) the perimeter of the union.

Input

The first line contains an integer n ($1 \leq n \leq 200000$).

Each of the next n lines contains four integers x_1, y_1, x_2 , and y_2 describing one rectangle ($-10^9 \leq x_1 < x_2 \leq 10^9$ and $-10^9 \leq y_1 < y_2 \leq 10^9$).

Output

Print two integers: the total area covered by exactly one rectangle and the perimeter of the union, in that order.

Both values fit in a signed 64-bit integer but may exceed the signed 32-bit range.

Sample input 1	Sample output 1
2 0 0 2 2 1 0 3 1	4 10

Notes

Rectangles are half-open only to make point membership unambiguous. Changing which rectangle owns a shared boundary does not change either requested value. Use 64-bit integer arithmetic.

Problem L – Ledger of true names

In the ruined observatory above the dead city, Racso and Christian copy out the Ledger of True Names.

A true name is power: speak one and the thing it belongs to must answer. But a name spoken alone burns the speaker, so the old sorcerers bound names in **pairs**, sealing each pair under a single sigil. The binding holds only if the two names **begin the same way**, and the sigil’s power is exactly the length of that shared beginning.

The brothers have N true names before them and one night of candlelight. Find the power of the strongest sigil they can seal — the longest beginning shared by **two different names** in the Ledger.

Input

The first line contains a single integer N ($2 \leq N \leq 10^5$) — the number of true names in the Ledger.

Each of the next N lines contains one true name s_i , a non-empty string of lowercase Latin letters with $|s_i| \leq 50$.

It is guaranteed that the sum of $|s_i|$ over all names does not exceed 10^5 . Names are not necessarily distinct.

Output

Print a single integer: the power of the strongest sigil, or 0 if no two names in the Ledger even begin with the same letter.

Sample input 1 5 racso rascal christian christ chris	Sample output 1 6
Sample input 2 3 umbra umbra sol	Sample output 2 5

Notes

Two names s and t share a beginning of length ℓ if $s_1 = t_1, s_2 = t_2, \dots, s_\ell = t_\ell$.

Two entries of the Ledger are always considered different if they sit at different positions in the input, even when their names are identical: two entries reading **umbra** share a beginning of length 5.

Problem M – Miguel and His Chess Tournament

Miguel organized a chess tournament with n participants numbered from 1 to n . Every pair played exactly one game. Every game ended with one winner and one loser; there were no draws.

Everything was going well until the software recording the results broke down. The individual game results disappeared, and only a table of totals could be recovered: according to it, participant i earned exactly b_i victories.

Miguel and the participants do not trust this table. If the program lost the games, who can guarantee that the totals were not corrupted too? Before announcing the winner, Miguel wants to know whether those numbers could describe a tournament like the one they organized.

Determine whether there is a set of game results consistent with the recovered totals. If there is, reconstruct one. You do not need to recover the original games: any reconstruction in which each participant i won exactly b_i times is acceptable.

Input

The first line contains an integer n ($1 \leq n \leq 2000$).

The second line contains n integers $b_1, b_2, \dots, b_n$ ($0 \leq b_i \leq n - 1$): the recorded number of victories for each participant.

Output

If no valid reconstruction exists, print NO.

Otherwise, print YES, followed by n lines, each containing a string of exactly n characters, all either 0 or 1, without spaces. These strings form a matrix A , where $A_{i,j} = 1$ means that participant i defeated participant j .

The matrix must satisfy $A_{i,i} = 0$, $A_{i,j} + A_{j,i} = 1$ for all $i \neq j$, and $\sum_{j=1}^n A_{i,j} = b_i$ for every i . If several valid reconstructions exist, print any of them.

Sample input 1 4 0 0 3 3	Sample output 1 NO
Sample input 2 1 0	Sample output 2 YES 0
Sample input 3 3 2 0 1	Sample output 3 YES 011 000 010

Problem N – Nord’s Cairn

Deep beneath the frost of Hjaalmarch, the excavation at **Ustengrav** has unearthed the collapsed remains of ancient Nord **burial cairns** — stone monuments the old draugr-lords raised over their fallen thanes.

The Nords built every cairn the same way: the topmost layer holds a single stone, and each layer beneath it holds exactly one stone more than the layer above. A pile of stones is called **cairn-perfect** if its stones can be arranged into such a cairn, using every stone and leaving no layer unfinished. Equivalently, a positive integer is cairn-perfect exactly when it is a triangular number

$$T_k = 1 + 2 + \cdots + k = \frac{k(k + 1)}{2}$$

for some integer $k \geq 1$.

Urag gro-Shub, keeper of the Arcanaeum at the College of Winterhold, has grown impatient with the excavation ledgers. The diggers record only the number of stones in each rubble heap, never its shape, and Urag will authorise no further shipment until he knows how many of the recorded counts could possibly have been complete cairns.

Given two integers L and R , determine how many integers n with $L \leq n \leq R$ are cairn-perfect.

Input

The single line of input contains two integers L and R ($1 \leq L < R \leq 10^{12}$), the smallest and largest stone-counts recorded in the excavation ledger. Both bounds are inclusive.

Output

Print a single integer: the number of cairn-perfect stone-counts in the ledger’s range.

Sample input 1 1 10	Sample output 1 4
Sample input 2 6 9	Sample output 2 1